Microbiology 101 – course syllabus

Monday-Wednesday-Friday: 10:30am-12:30pm

Classroom: - / Laboratory: -

Who will be teaching?

My name is Cilicia Nascimento. I come from the beautiful country of Brazil, I was born in our largest city, Sao Paulo, so I am used to a chaotic life. When I was in undergrad and took an introductory microbiology course, I did not know I would eventually get a PhD in microbiology and come to America. During that course I had this feeling of "this is awesome, I need to know more!" It is my hope that, by the end of our course, you will share that same feeling. One of my favorite activities outside of the lab/classroom is crafting, and crochet is my current passion: I am proud to show you my latest creation. Having a hobby is very important and I hope we have time to share what our favorites are!

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Office hours: Monday 1-5pm and Friday 2-6pm

My teaching philosophy: As an educator, it is my responsibility to provide a safe environment for all students. In this course, you have the freedom to be whoever you are. All racial, religious, ethnic or gender profiles are accepted and welcomed. Remember, this is a learning environment, so feel free to ask questions, propose discussions or activities. I believe that learning is an on-going process, and all input regarding how this course is conducted is welcomed, I am always looking to improve my methods. I believe that each student has a different learning method that is optimal for them, I am here to provide tools that will work best for each of you.

What is this course about?

In this course we will explore the micro-world. Be ready to answer important questions, such as: What are bacteria? Do we really have more microorganisms in our body than human cells? What is the difference between bread mold and mushrooms? What is a pathogen? Why do we get food poisoning?

We will have a very inclusive approach, so there is no need to worry about having a science background. But I believe that understanding scientific literature is an important skill, so we will learn how to approach a scientific paper and discuss relevant articles at the end of every unit (check "Unit Science time!" on our calendar below). By the end of this course, microbiology will no longer be a mystery, and you will be ready to join scientific discussions about important microbial topics in today's world.

By the end of this course, you should be able to:

- Apply multiple techniques, including both microscopy and biochemical tests, to identify several microorganisms, and provide rationale for every reaction or test

- Understand how microbiology concepts apply to our daily lives (for example, why is it important to thoroughly wash our hands but we also tell children to not be afraid to play in the dirt?)
- Engage in discussions about scientific literature on the topic of microbiology (why is it important to share about bacteria resistant to multiple antibiotics?)
- Know the importance of the human microbiome and apply that knowledge to your routine
- Interpret microbiology-related news
- Attend and succeed in advanced microbiology classes

How do we achieve those goals?

- Following the lectures and taking full advantage of the accompanying textbooks and supplemental materials will ensure optimal comprehension of all the content covered in this course
- We will have several activities that will assess your knowledge regarding what was covered in class, so completing them is essential for the development of your understanding
- Have a question? Always ask! This is a learning environment, so all questions are welcome, both during class and during my office hours
- I will give feedback on all activities to help guide your microbiology-learning journey
- A teacher's assistant will be present in class during our reviews at the end of each unit, they will also be available during office hours, so feel free to take your questions to them

Our learning resources:
Main textbook: "Microbiology: an introduction"; Tortora, Gerard J., Berdell R. Funke, and Christine L. Case. Published by Pearson. 13th or 14th ed.
Auxiliary textbook: "Brock Biology of Microorganisms"; Madigan, Michael T., Bender, Kelly S., Buckley, Daniel H., Sattley, Mathew W., Stahl, David A. Published by Pearson. 15th or 16th ed.

Grading:

Your grade will have three components:

- **Class activities:** we will have one presentation and one activity that should be performed in groups. For these activities, groups will grade each other. I will evaluate the grades and attribute a grade according to your feedback.
- **Quizzes:** at the end of every unit, we will have a quiz that will be taken online, due one week after the last class of that unit.
- **Final assignments**
    - **Content assignment:** this assignment must show what you learned throughout this course. The form this assignment will take is chosen by each student. Some options are essay, presentation, video, lecture, quiz and oral quiz. All final assignments are due on the day appointed in the calendar below. In the case of presentations or lectures, those can be given in class on that day.

- **Self-assessment:** one week before the due date of the final assignment, I will have the "self-assessment" form available to be filled out online. The form will cover how this course impacted your knowledge on the subject, and how it will affect your career moving forward.

IMPORTANT: The weight of each grade component will be determined by each student.

Weight options are:

a) **50-20-30**
b) **60-20-20**
c) **50-25-25**
d) **40-30-30**

For example, you chose option c), and decided to distribute the weight as follows:

Class activities: 50%

Quizzes: 25%

Final assignments: 25%

So that should be the components and weight of your final grade.

Due dates

All activities are due one week after their assignment. The dates are marked in the calendar, and I'll communicate any changes in advance. The knowledge acquired through such activities is essential for a better understanding of the following lectures, so their non-completion or delay can create gaps in knowledge. However, I will be accepting quizzes with no grade deduction up to a week after the due date. I cannot accept submissions any later than that to avoid accumulating a large volume of corrections by the end of the semester. For the in-class activities, please let me know if you won't be able to attend or if a reschedule is needed, I will try my best to adjust accordingly.

About "Unit Science time!"

Science is not static; it is constantly evolving. One way that we scientists keep up with what is happening in labs all over the world is through publications in scientific journals. In this course we will not only learn the history of microbiology and the important discoveries of the past, but we will also keep up with the latest research. We will do that through the "Unit science time!", which happens at the end of every unit. I will choose one (or more) recent scientific publications for us to read, understand, and discuss in class. What experiments were done? Are the conclusions coherent with the results that were obtained? Reading, discussing and understanding scientific publications is a valuable skill that will help you in your academic and professional journey, so this will be a great opportunity to develop that skill.

Calendar:

Date	Class title	Class summary	Assignments
Unit 1 – A microbe introduction: Here we get familiarized to the concept of "microbiology" and learn about important discoveries that were made in the field throughout the years.			
	What is microbiology? - A brief history of germs	We will talk about the discovery of microorganisms and draw a timeline from the germ theory until today.	
	Under the microscope	We will understand the principles and components of the microscope, a brief history of microscopy. I will go over types of microscopes and we will learn about several different stains. This lecture includes a **laboratory** activity.	
	Building bacteria	We will learn about the structure and function of several different bacterial components, like cell walls, membrane, flagella, and others.	
	HOLIDAY – Labor Day		
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	Quiz1: to be taken home I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
Unit 2 – Bacterial functions and identification: **Here we will learn how bacteria have been surviving and thriving in different environments and how we can identify a bacterial species based on what we know about them**			

	Bacterial reproduction and genetics	We will learn about bacterial replication, transcription and translation. We will also talk about regulation of expression and recombination.	
	Bacterial metabolism	We will learn how bacteria obtain energy; we will participate in an activity that illustrates the electron chain. And I will show how bacteria metabolize complex molecules.	**Quiz1: DUE DATE**
	DNA technology	We will go over the principles of different molecular biology techniques (like PCR, sequencing, cloning and others).	**Optional:** bring a sample you'd like to see in a petri dish
	Bacterial identification	This lecture will be in the **lab**. We will perform different biochemical tests and learn what is the principle behind each one.	A step-by-step **report** on how to identify a bacterium from the sample analyzed by your group. You should include what microorganism(s) is(are) present according to your results. And what molecular biology techniques could be used to confirm your results.
	Molecular biology techniques	This lecture will be in the **lab**. We will perform a PCR targeting specific bacterial genes and run an electrophoresis gel to visualize our results	
	Pathogenicity	We will learn about mechanisms used by	

		microorganisms to establish colonization and infection	
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	**Lab Report: DUE DATE** Quiz2: to be taken home. I will upload the articles that will be discussed in the next class
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
Unit 3 – Learning about specific microorganisms **Here we will learn the names and last names (genus and species) of important bacteria, and how they relate to human health and disease**			
	Gram-negative bacteria	We learned about Gram staining, now we will go over the main microorganisms that belong to the Gram-negative group.	In **groups**: each group will choose one Gram positive bacteria and develop an activity based on it. That should include what we learned about bacteria so far.
	Gram-positive bacteria	This lecture will be a collaboration between students. Each group will apply their activity to the whole class (including myself).	**Quiz2: DUE DATE** Grading each other's Gram-positive activity.
	Microbial diseases	Let's learn about some bacteria of medical importance. We will discuss diagnosis, symptoms and treatment. We will also talk about antimicrobial drugs.	We will talk about the assignment where you will prepare a lecture about two microorganisms of your choice (from a list). That lecture will be given in the next unit.
	The human microbiome, what we know so far	We will learn about the Human microbiome project,	

		main characteristics of microbiomes from several sites in the body. And how our knowledge about microbiomes is shaping modern medicine.	
	Epidemiology	We will understand how disease spreads and how it affects society.	
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	Quiz3: to be taken home. I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
	HOLIDAY: Columbus Day		
Unit 4 – Time to apply our knowledge! **In this unit we will dive into specific microbial infections.**			
	Microbial infections in humans: skin and eyes	During these, each lecture will be divided into 2 parts: in the first one I'll introduce the subject and talk about many examples of microbial diseases pertinent to the class, in the second part, one or more groups will give a presentation about one microorganism previously chosen.	Group presentation
	Microbial infections in humans: skin and eyes		**Quiz3: DUE DATE** Group presentation
	Microbial infections in humans: the cardiovascular and lymphatic systems		Group presentation
	Microbial infections in humans: the respiratory system	During these, each lecture will be divided into 2 parts: in the first	

	Microbial infections in humans: the digestive system	one I'll introduce the subject and talk about many examples of microbial diseases pertinent to the class, in the second part, one or more groups will give a presentation about one microorganism previously chosen.	Group presentation
	Microbial infections in humans: the urinary and reproductive systems		Group presentation
	Environmental microbiology		Group presentation
	Unit review	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	Grading each other's lectures. I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
Unit 5 – Broadening our horizons **We have explored the amazing microbial world with a special emphasis on bacterial human infections, now it is time to learn about viruses and fungi**			
	Introducing virology	We will discuss the main human viruses' infection mechanisms and discuss about what changed in our epidemiological approach since the SARS-CoV-2 pandemic.	
	Virus structure and replication	We will learn the mechanisms involved in viral reproduction	
	Human viral infections	We will discuss the most important viral infections and build a timeline of viral pandemics throughout history.	

	Environmental virology	We will learn about the viruses that inhabit our environment, and how humans have adapted to them and vice versa.	
	Introduction to Fungi	We will talk about the kingdom Fungi, that encompasses a multitude of organisms.	
	Fungi of medical importance	We will discuss a few fungal infections, including their symptoms and treatment.	
	Fungal Metabolism	We will learn the unique mechanisms involved in how fungi obtain energy and thrive in different environments.	
	Environmental importance of Fungi	We will go over how viruses can influence the environment and the latest biotechnological advances in fungal science.	
	Unit review and quiz	We will go over the main topics of the previous unit, and we will answer all questions and take a closer look at concepts that might still be cloudy.	Quiz4: to be taken home. I will upload the articles that will be discussed in the next class.
	Unit Science time!	We will bring one (or more) scientific articles that are pertinent to this unit to be discussed in class	
	HOLIDAY - Thanksgiving		
	Mega review	We will go over the burning questions	**Quiz4: DUE DATE**

		regarding all topics we went through this semester. We will have a review activity that does not account for grading. And I will explain what the final assignment will look like and what I expect from it.	
	Final assignment	Each student will take their final assignment as they chose previously. The final self-assessment will be filled out online.	
	Last day of class!	I will make your final grade available for you one day before our last day of class. And I will be available in class during our last day to discuss grades and answer questions.	